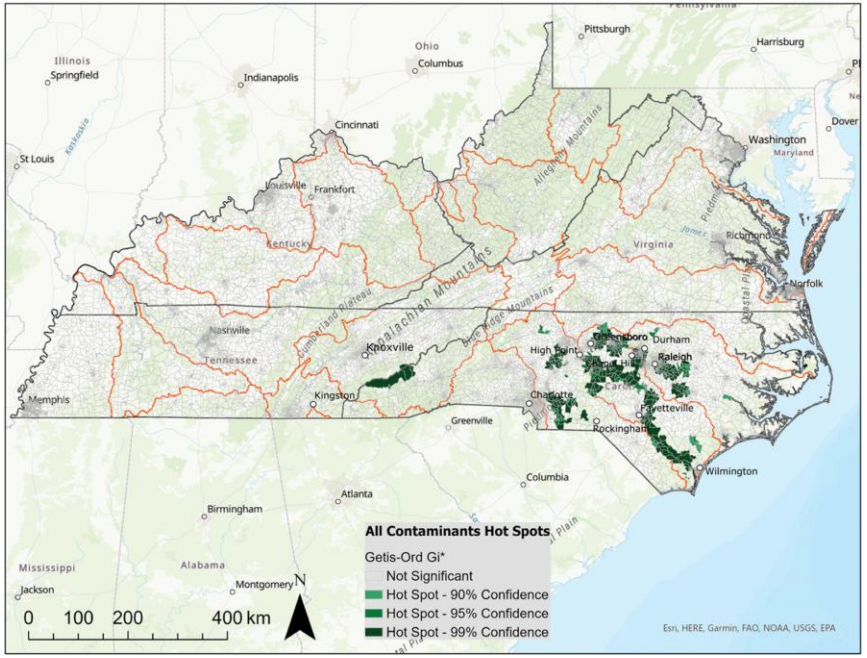
Lecture 5: Spatial statistics

- Spatial sampling design.
- Central tendency, dispersion and direction.
- Inferential statistics and spatial randomness.
- Spatial autocorrelation.
- Spatial clusters analysis.
- Nearest Neighbour, Moran's I statistic (Cluster/Outlier), Getis-Ord (Hot Spot) and Kernel density estimation methods.

Spatial statistical analysis

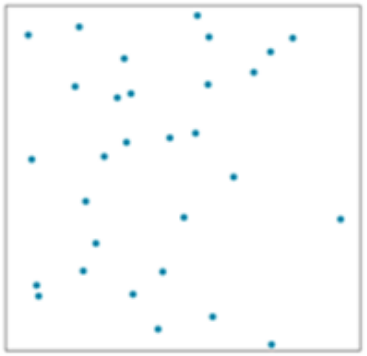
Applying spatial statistical methods and tools to spatial data.

- Minimise the subjectivity inherent in human visual interpretation of maps and spatial data.
- Identify and quantify patterns and trends in data that may not be revealed in visual analysis.
- Answer questions more confidently and make important decisions using more than simple visual analysis.



Hotspot analysis results for mercury, lead, copper, zinc, nickel, and fluoranthene within the central Appalachia region (Neville et al. (2022)DOI: 10.1080/02626667.2022.2052073

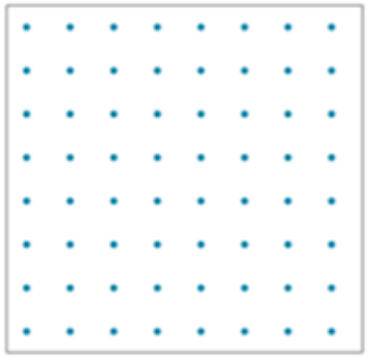
Spatial sampling methods



Random



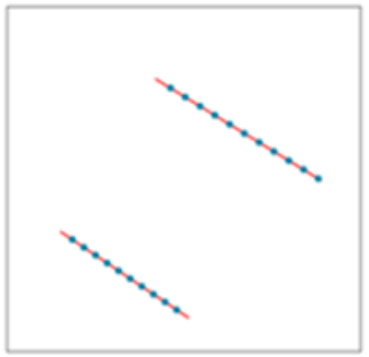
Stratified



Systematic



Clustered

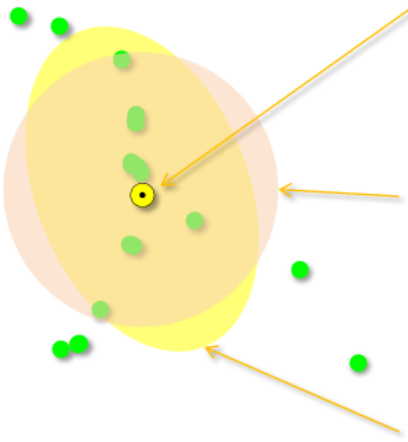


Transect

Descriptive statistics

- Descriptive statistics summarise data (quantitative, visual).
- Commonly measure central tendency, dispersion or concentration, and orientation of spatial phenomenon.
- Central tendency
    - *describe the centre of a spatial distribution*.
- Dispersion
    - *describe the degree of dispersion of geographical features around the centre of their spatial distribution*.
- Direction
    - *describe directional spread of data (e.g. wind directions, animal migrations, pollution pathways, volcanic plumes, etc)*.

Measures of central tendency and dispersion



Mean center
computed average X and Y coordinate values.

$$\bar{s} = \left(\frac{\sum_{i=1}^{n} x_i}{n}, \frac{\sum_{i=1}^{n} y_i}{n} \right)$$

Standard distance
measure of the variance between the average distance of the features to the mean center.

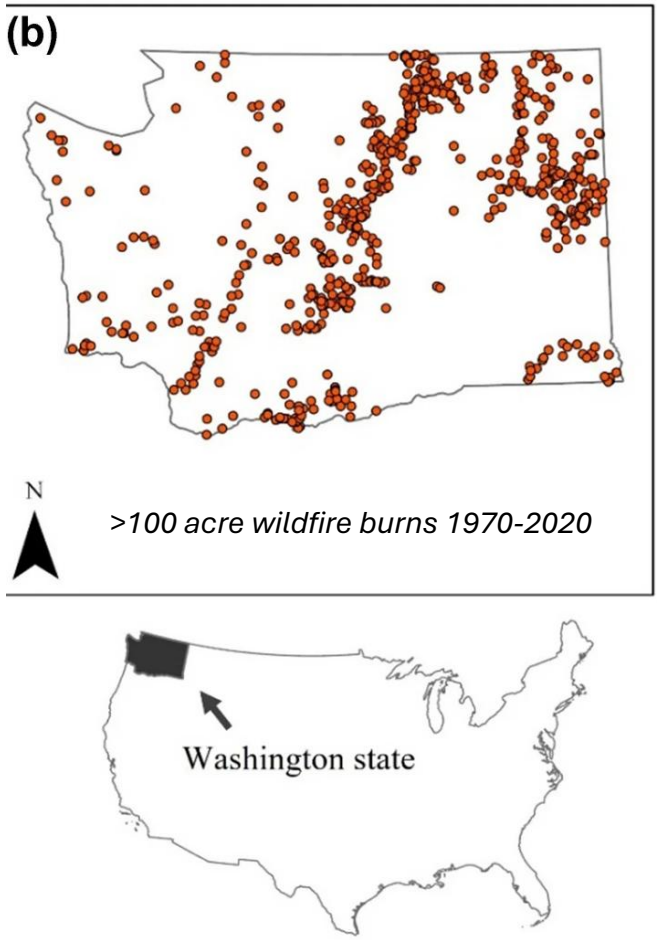
$$d = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu_x)^2 + (y_i - \mu_y)^2}{n}}$$

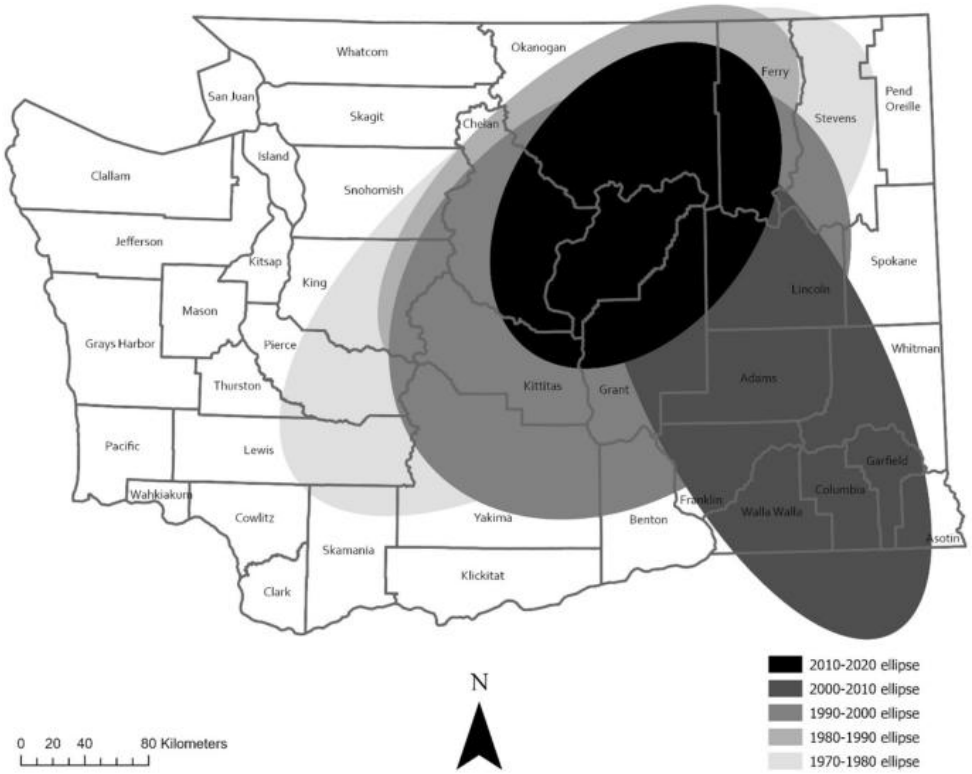
Standard deviational ellipse
separate standard distances for each axis.

$$d_x = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu_x)^2}{n}}$$

$$d_y = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \mu_y)^2}{n}}$$

Directional distribution



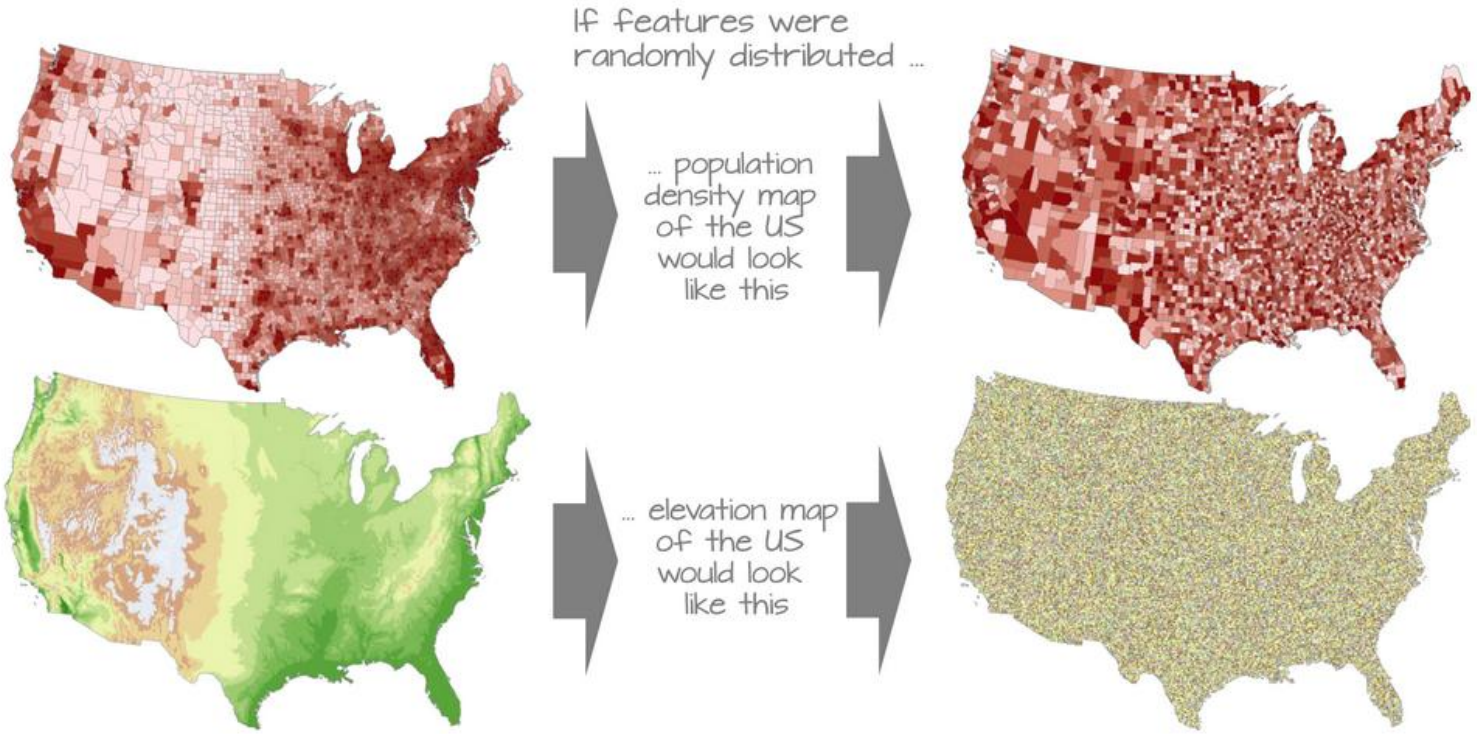


https://link.springer.com/article/10.1007/s13753-022-00396-4

Inferential statistics

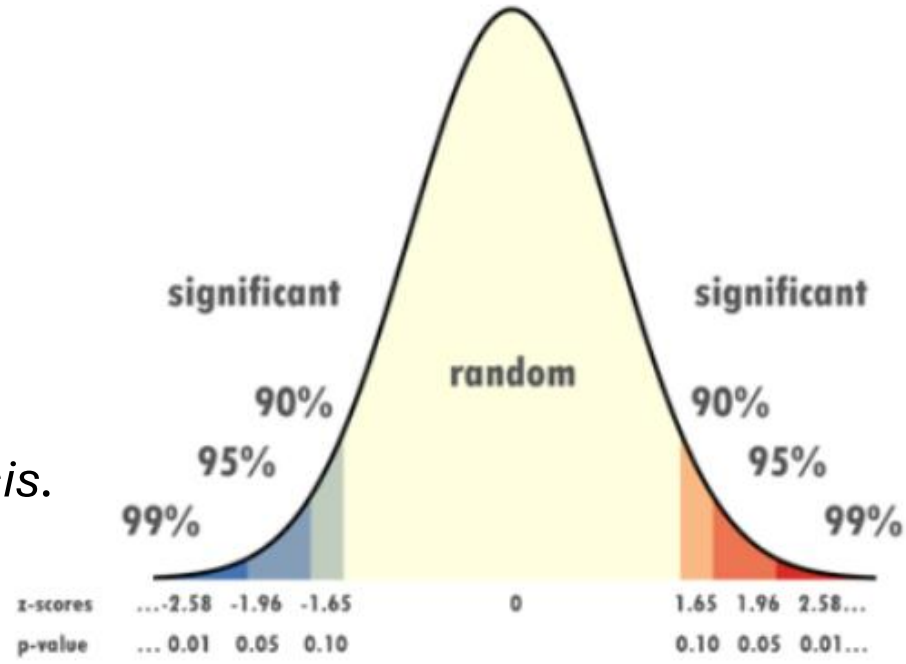
- Inferential statistics means to infer something about the entire population based on the distribution of values in a sample.
- Statistical significance is the probability that the observed spatial pattern is not likely due to chance alone (random).
  - *The observed spatial pattern is considered significant if it has been predicted as unlikely to have occurred by chance.*
- Significance or hypothesis testing.
  - *Determine whether the spatial pattern quantified by a statistical measure could have arisen by chance alone or through sampling error.*

Complete spatial randomness



Null and alternative hypotheses

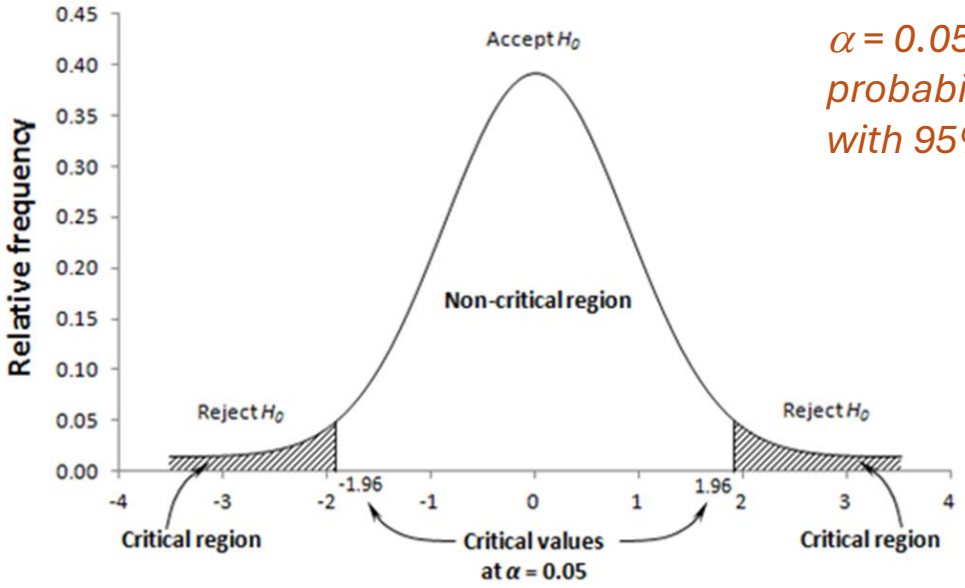
- Null hypothesis (H_0).
  - ***always*** *stated as being no spatial pattern, or no relationship between two observed spatial patterns.*
- Alternative hypothesis (H_a).
  - *opposite to the null hypothesis.*
- Significance test – accept or reject the null hypothesis.



Standard normal distribution for randomness

z-scores and p-values

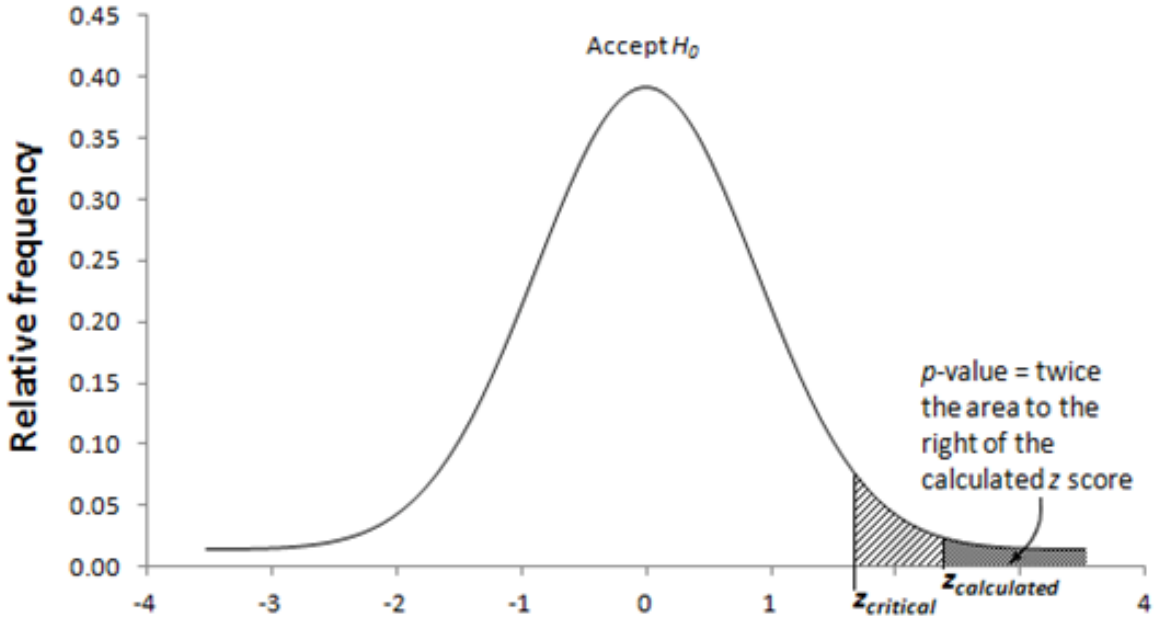
- z-score represents a deviation from the mean in a distribution.
- A p-value (confidence level) dictates the critical z-score for rejecting or not rejecting the null hypothesis (H_0).



$\alpha = 0.05$ means a probability of rejecting H_0 with 95% confidence.

z-values

Finding *p*-values in the probability distribution of the *z* scores



The exact probability of getting a value of the test statistic of a given magnitude if the null hypothesis is true is called the *p-value*

Nearest neighbour analysis (NNA)

- Measure the distance of each feature to its nearest neighbour (r_0) and compare the observed distances with that expected from a random spatial pattern (r_e).
- Uses only feature location and **does not** consider the feature values at all.
- Ratio (R) = mean observed/ mean random distances.

  R < 1 then tendency toward clustering

  R = 1 then tendency to be random

  R > 1 tendency to be dispersed.



Q. Are these sample locations random located from eachother?

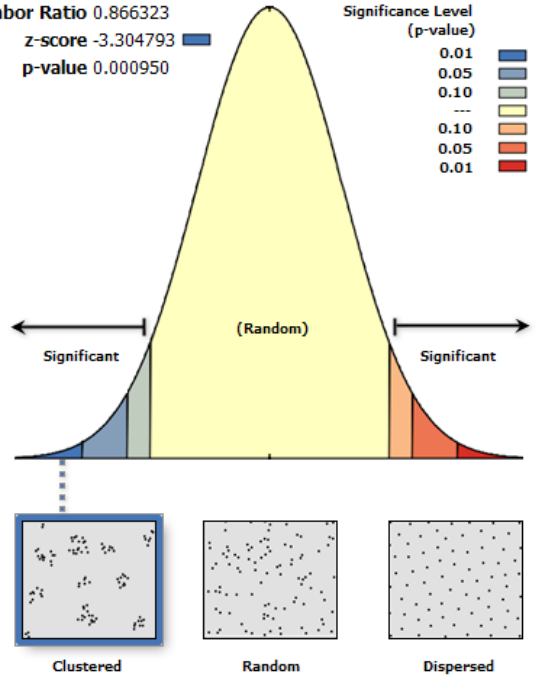
Nearest neighbour analysis



Average Nearest Neighbor Summary

Nearest Neighbor Ratio 0.866323
z-score -3.304793
p-value 0.000950

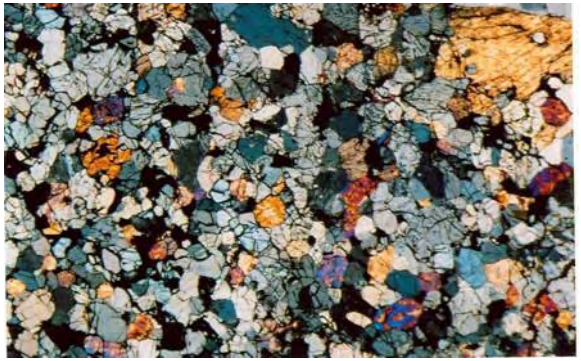
Significance Level (p-value)	Critical Value (z-score)
0.01	< -2.58
0.05	-2.58 - -1.96
0.10	-1.96 - -1.65
---	-1.65 - 1.65
0.10	1.65 - 1.96
0.05	1.96 - 2.58
0.01	> 2.58



Given the z-score of -3.304793, there is a less than 1% likelihood that this clustered pattern could be the result of random chance.

NNA examples

Mineral textures



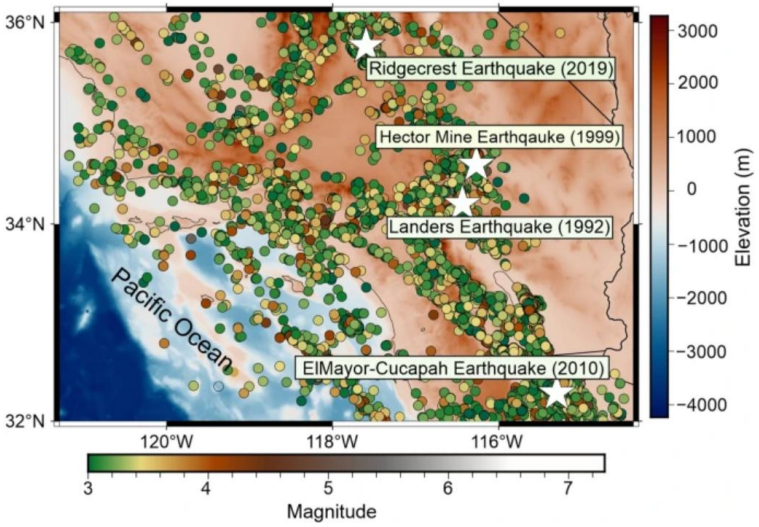
Tree surveys





Paleoecology

Earthquakes



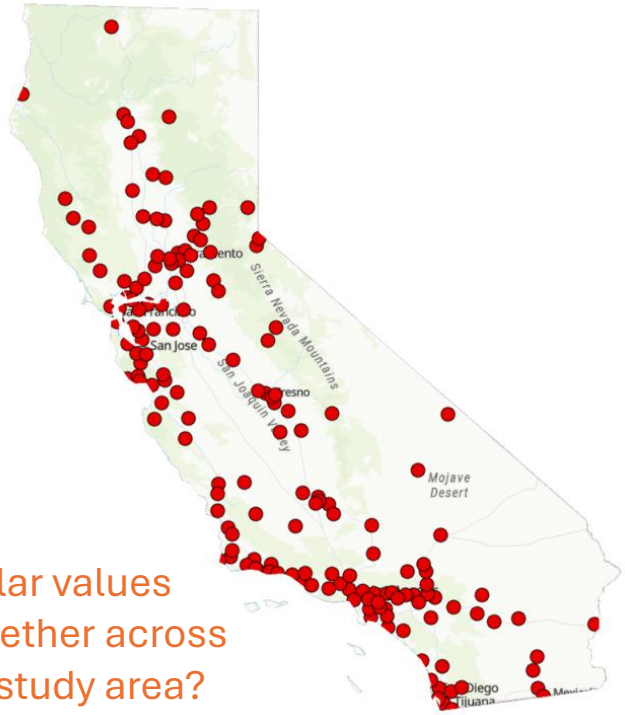
Spatial Autocorrelation

- Distance methods using both feature locations and feature values simultaneously.
- Evaluates the spatial autocorrelation.
- Determines if pattern is clustered, dispersed, or random.

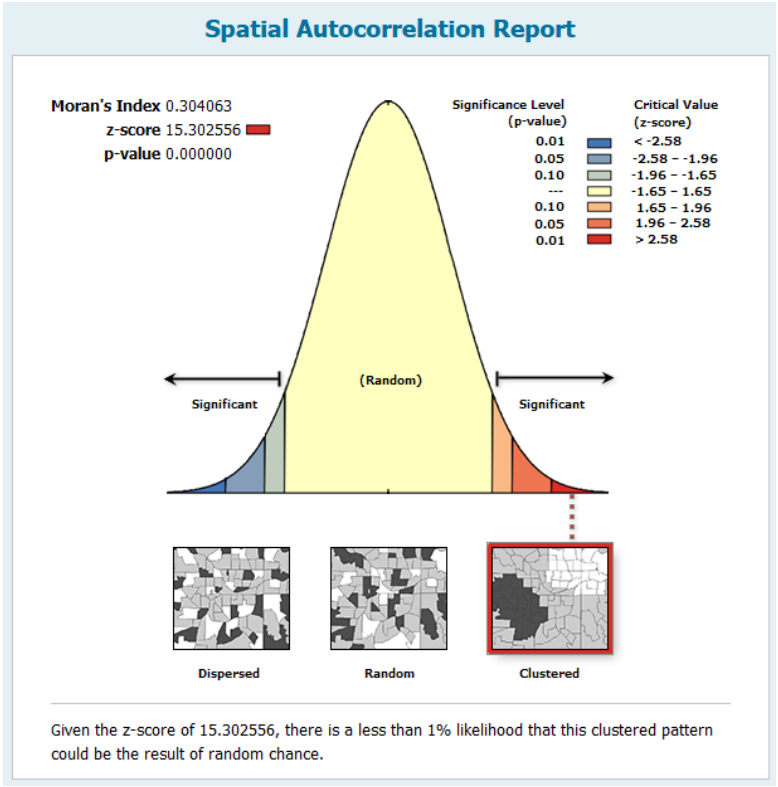


Global Moran's *I*

Detects spatial clusters of features with high or low values (and spatial outliers). *Works with attribute values*.

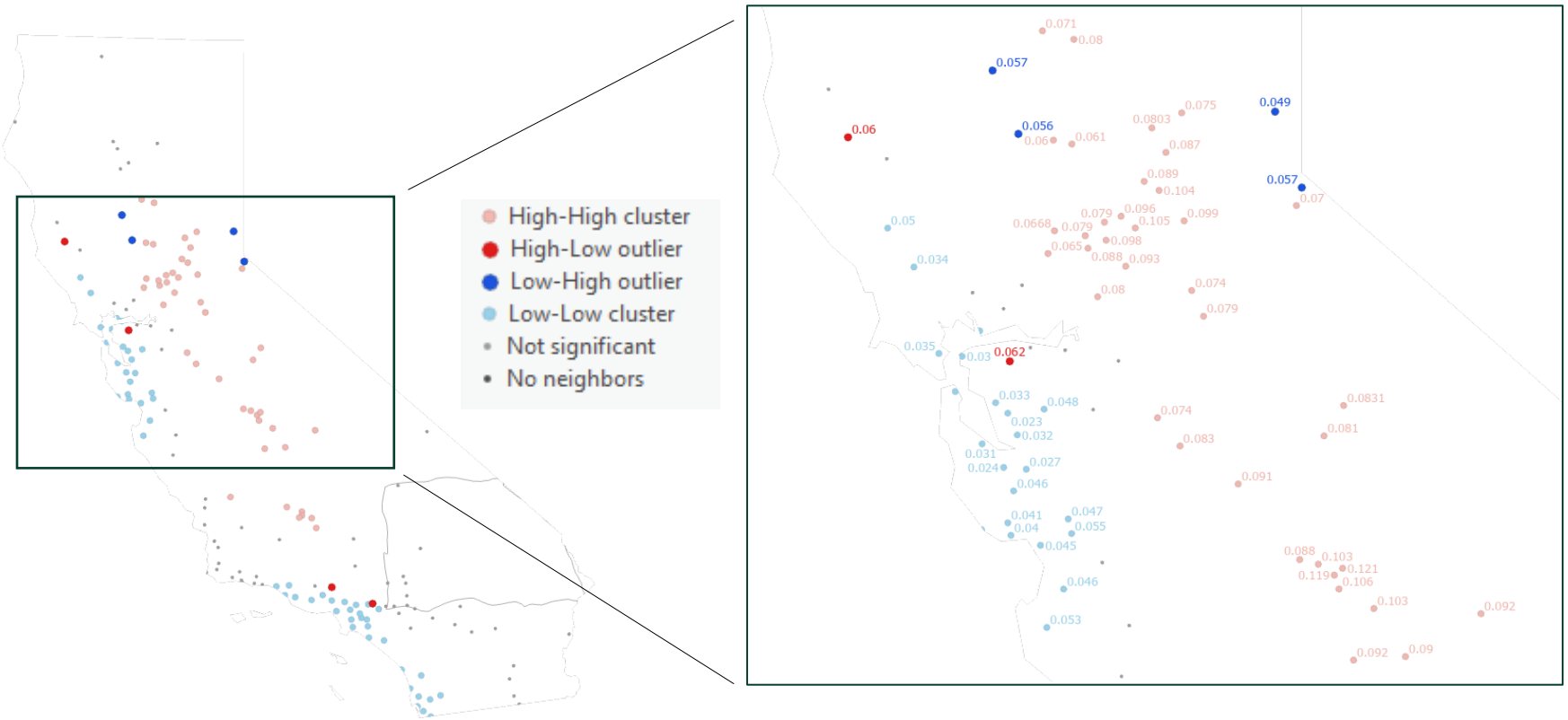


Q. Do similar values cluster together across the whole study area?



Local Moran's *I*

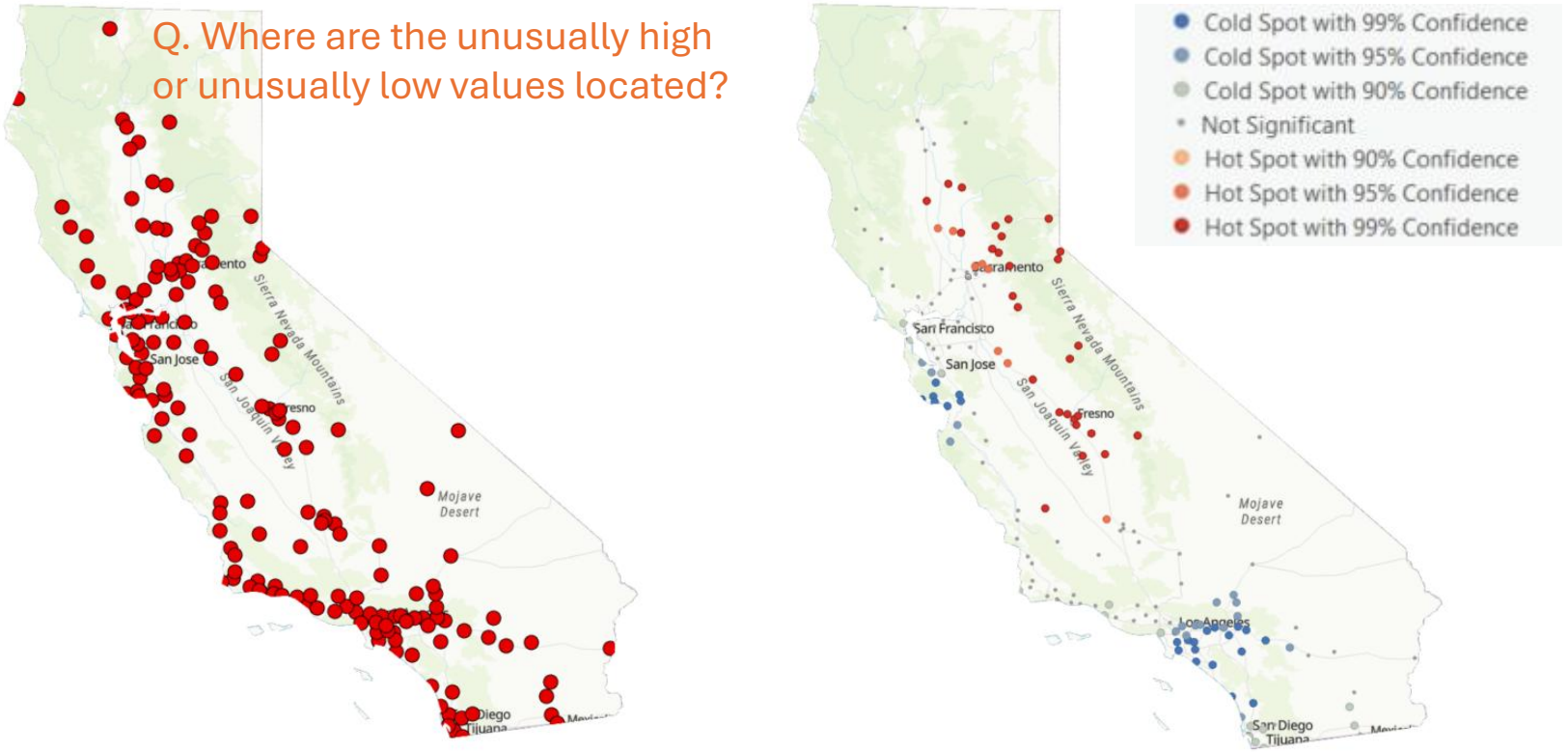
- Local Moran's I shows where clusters occur and identifies local outliers.



Mashhoodi (2024) https://doi.org/10.1007/s12061-023-09555-6

Hot spot analysis (Getis-Ord Gi*)

Q. Where are the unusually high or unusually low values located?



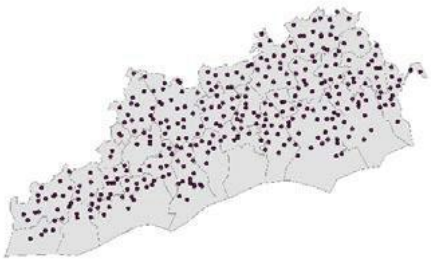
The hot spots (red) indicate statistically significant clustering of higher values. Cold spots (blue) are clusters with low values. *Works with attribute values*.

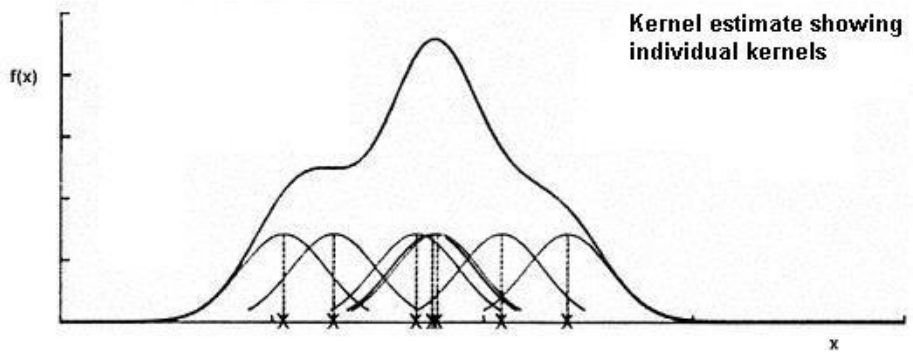
Kernel density estimation (KDE)

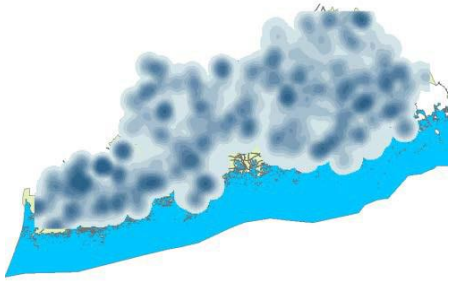
- Transformation from point objects to a density surface to visualize a point pattern to detect hot spots where the local density is estimated to be high.
- Density at any location in the study area.
  - estimated by counting the number of objects (or events) in a region (=kernel) centred at the location p where the estimate is made
  - simple kernel: circle centred at the location p.
  - more sophisticated: a kernel function weighting nearby events.
- Density is scale dependent.

Kernel density surface

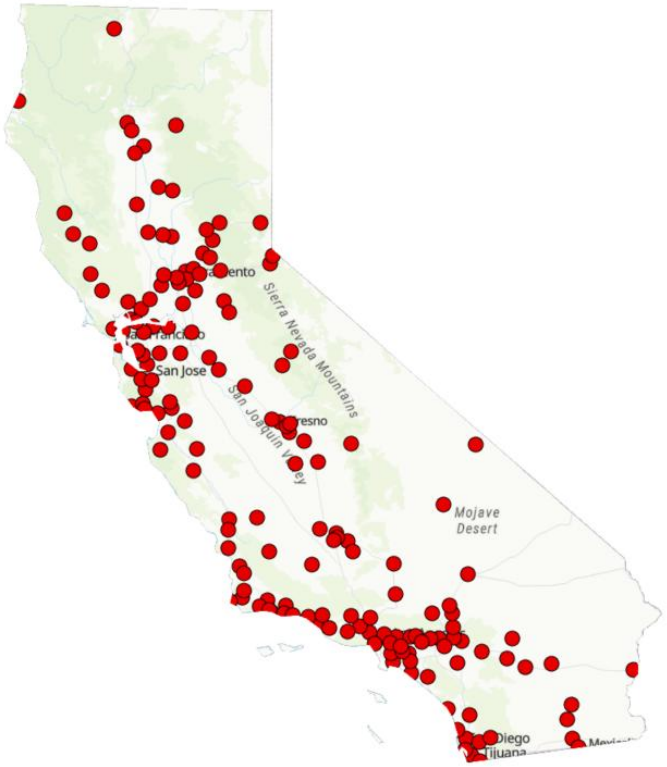
- Kernel function (e.g., bell shape) for the local density.
- Each point is replaced by a kernel.
- Density surface is the sum of these.
- Density surface estimates the variation of the density of discrete events (events = point objects).

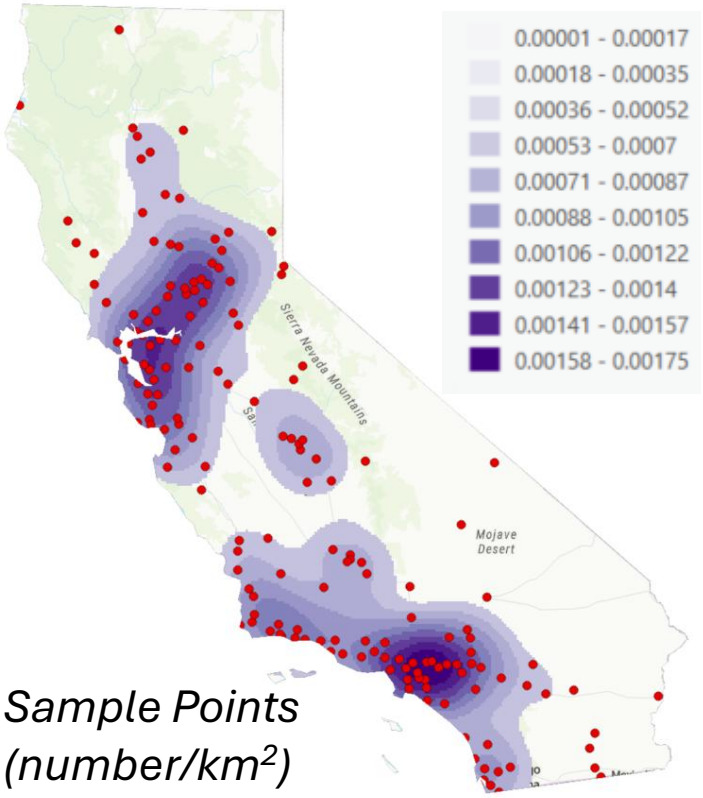






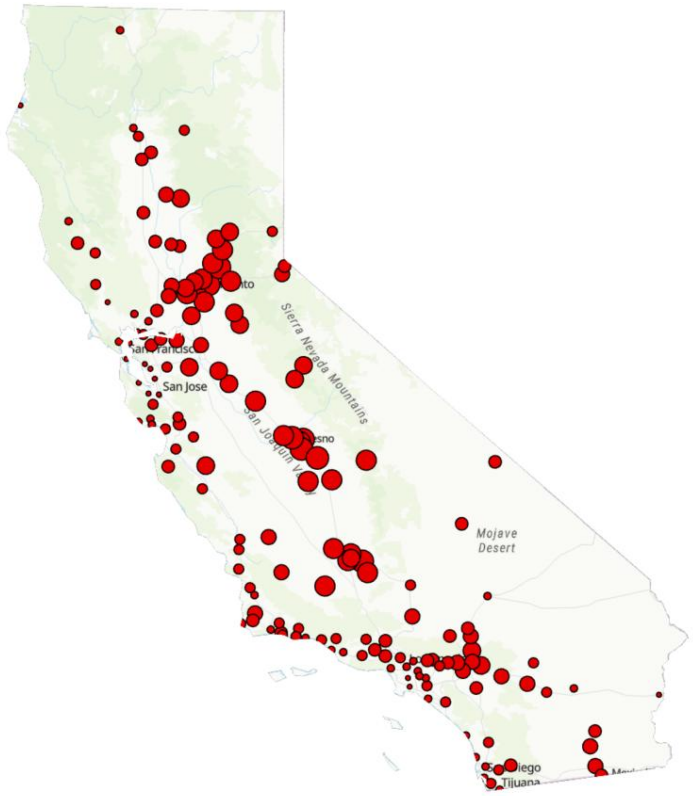
Heat map using kernel density: sample point

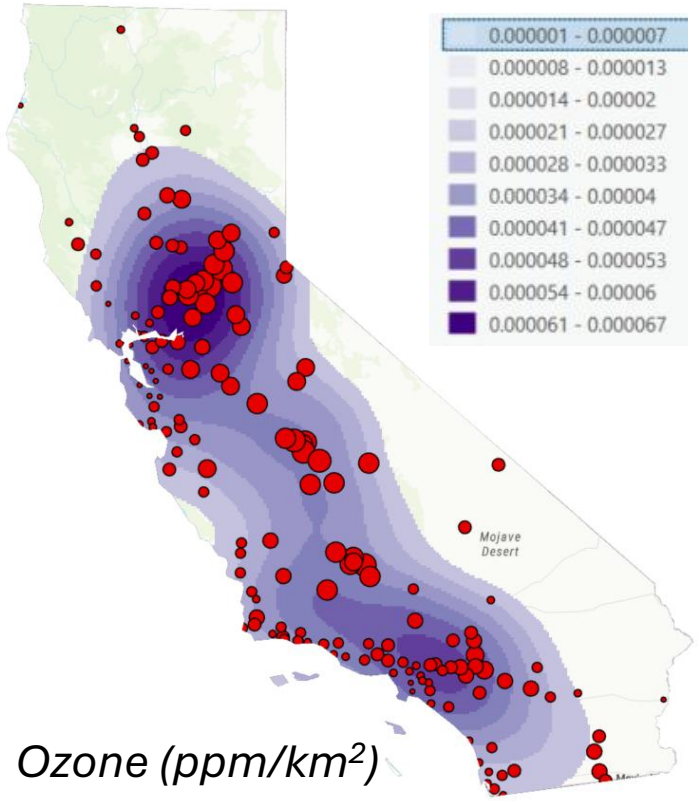




Sample Points (number/km^2)

Heat map using kernel density: ozone

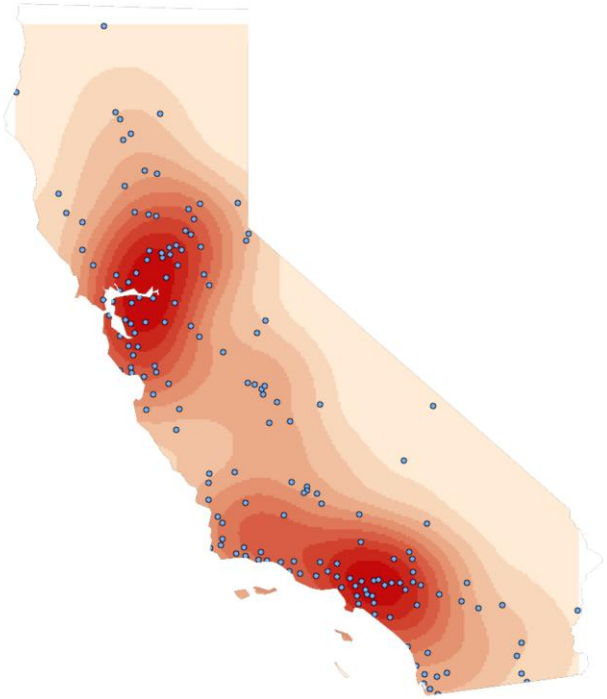




Ozone (ppm/km^2)

Kernel radius (bandwidth)

1500 km



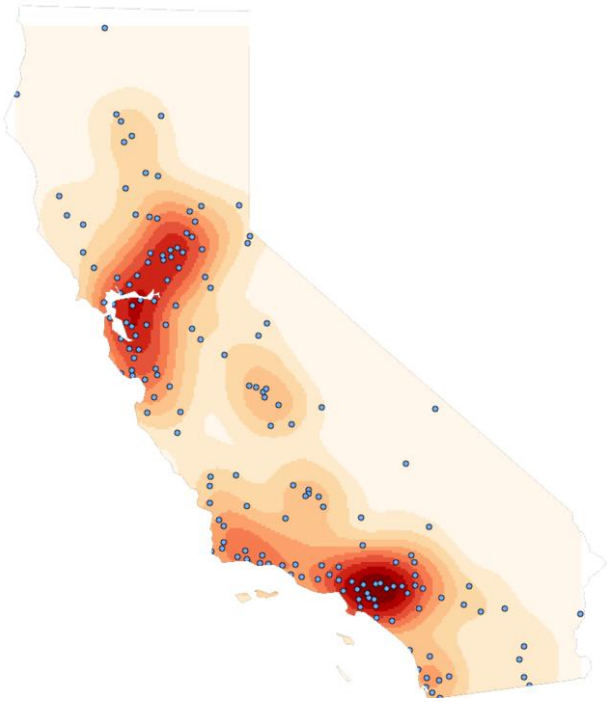
Smoother (global average)

15km



Isolated on individual values

optimised 99.3 km



Homework and Resources

- Read Chapter 6 of course textbook: Spatial Data Exploration with Statistics (you do not need to learn any statistical formulae).
- Take the quiz to test your basic understanding (on Canvas).

Further resources

ESRI training: Mapping Clusters: Introduction to Statistical Cluster Analysis (50 mins).

https://www.esri.com/training/catalog/5d39b6106b9c17664bb656eb/mapping-clusters-introduction-to-statistical-cluster-analysis/